3711.TW - ASE Technology Holding | Full investment artifact

Full investment research | Model assumptions and source evidence

Research date: 12 October 2026 | **Framework:** 8.0 consolidated | **Format:** standalone current-state analysis; reported Q3 sales calibrated; pricing branch retained **Status:** NOT HELD / screen in supplied original | **Classification:** BOTTLENECK - qualified advanced packaging and test capacity **Artifact Base:** 46/60 = **77% - High Conviction** | **Protected portfolio Base:** 77% (user override retained; separate from artifact assessment) **Snapshot:** supplied-workbook price TWD 585.000; model share proxy 4,540.633558M; implied snapshot market capitalization 2,656,270.631 TWD millions. Price is frozen from the supplied workbook, not a current-market refresh; capitalization uses the modeled denominator. **Active price target:** Q4 2027 **TWD836.638488** remains unchanged, using Q4 2027-Q3 2028 modeled operating profit. Reported Q3 sales replace the former guide midpoint; the Q3 2026 ceiling moves from TWD572.171631 to **TWD580.958890 (+1.535773%)**. Product mix, GP and OP remain estimates. The 20x method, shares, FX and both Base77 scores remain intact.

ASE's opportunity is expanding from more AI chips to more packaging and testing work per chip. Current margin gains and larger capacity commitments provide a firmer starting point. The full model carries that demand forward through qualified throughput and process content, while charging manufacturing costs, operating investment and known conversion dilution.

1. Supercycle, supply chain and durable advantage

ASE supplies constrained, qualified packaging and test capacity. The strongest evidence is rising ATM profitability alongside expansion commitments. Its independent OSAT role supports collaboration across architectures; the economics remain capital intensive. The conditional price target assumes new capacity achieves yield and customer qualification on time, rather than capitalizing every new technology as separate revenue.

DB5 advanced packaging and DB10 inference: accelerators/custom silicon and chiplets → substrate/interposer assembly, full-process packaging, wafer sort and final test → deployable computing systems. ASE's scale, process history and customer qualification support a bottleneck role. Foundries and rival OSATs remain alternatives. New packaging architectures change who performs each process; they are not automatically additive end markets.

Business / curve	Stage	Evidence and next inflection
CoWoS-adjacent work and test	Commercial ramp;10-50% qualitative	Current demand exceeds readily available qualified capacity.
Full-process and advanced test	Early ramp	Larger package area and longer testing increase work content.
Panel packaging and optical integration	Pre-ramp to early ramp	Pilot performance and qualified production govern timing.

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

2. Forward TAM and the three-phase expansion lens

Phase	Engineering opportunity	Demand-to-company bridge
Phase 1;100% thematic	Existing AI packaging and testing	Management's 2027 doubling objective is anchored in customer demand and planned execution. The 7.4B model translates an illustrative 3.7B 2026 outcome into twice the sales.
Phase 2; 50% thematic	More process steps, larger packages and full-process outsourcing	20B 2030 LEAP case requires 2× productive volume and 1.35135× invoice content versus 2027. Neither factor is an externally verified capacity forecast. This expands the opportunity beyond a fixed share of today's wafer volume.
Phase 3;20% thematic	Panels, optical/power integration and heterogeneous physical-AI devices	These broaden future package design and test needs. Revenue stays inside LEAP or mainstream ATM; no full CPO, chiplet and panel TAMs are summed.

The 100% / 50% / 20% thematic weights guide thesis assessment. They are not blanket discounts to reported sales, contracts or all forecasts. Programme probabilities below govern only separately identified optionality. Current reported sales calibrate the starting point; they do not cap the future market.

3. Current earnings and management evidence

Q2 2026 consolidated revenue was NT$191,064M, gross profit NT$40,150M and operating profit NT$21,134M. ATM revenue was NT$126,148M at 27.3% GM; EMS revenue NT$65,789M at 8.9% GM. Consolidated revenue includes intercompany eliminations. Latest diluted weighted-average shares: 4,505.702583M. July and August sales were NT$73,784M and NT$82,247M.

Reported September and Q3 sales: current-quarter calibration

The October12 issuer release reports unaudited September consolidated sales of **TWD88,130M** and Q3 sales of **TWD244,161M**. Q3 ATM sales are **TWD151,485M**. The remaining **TWD92,676M** is the derived non-ATM/consolidation residual; it is not a separately reported standalone EMS revenue figure. [A10]

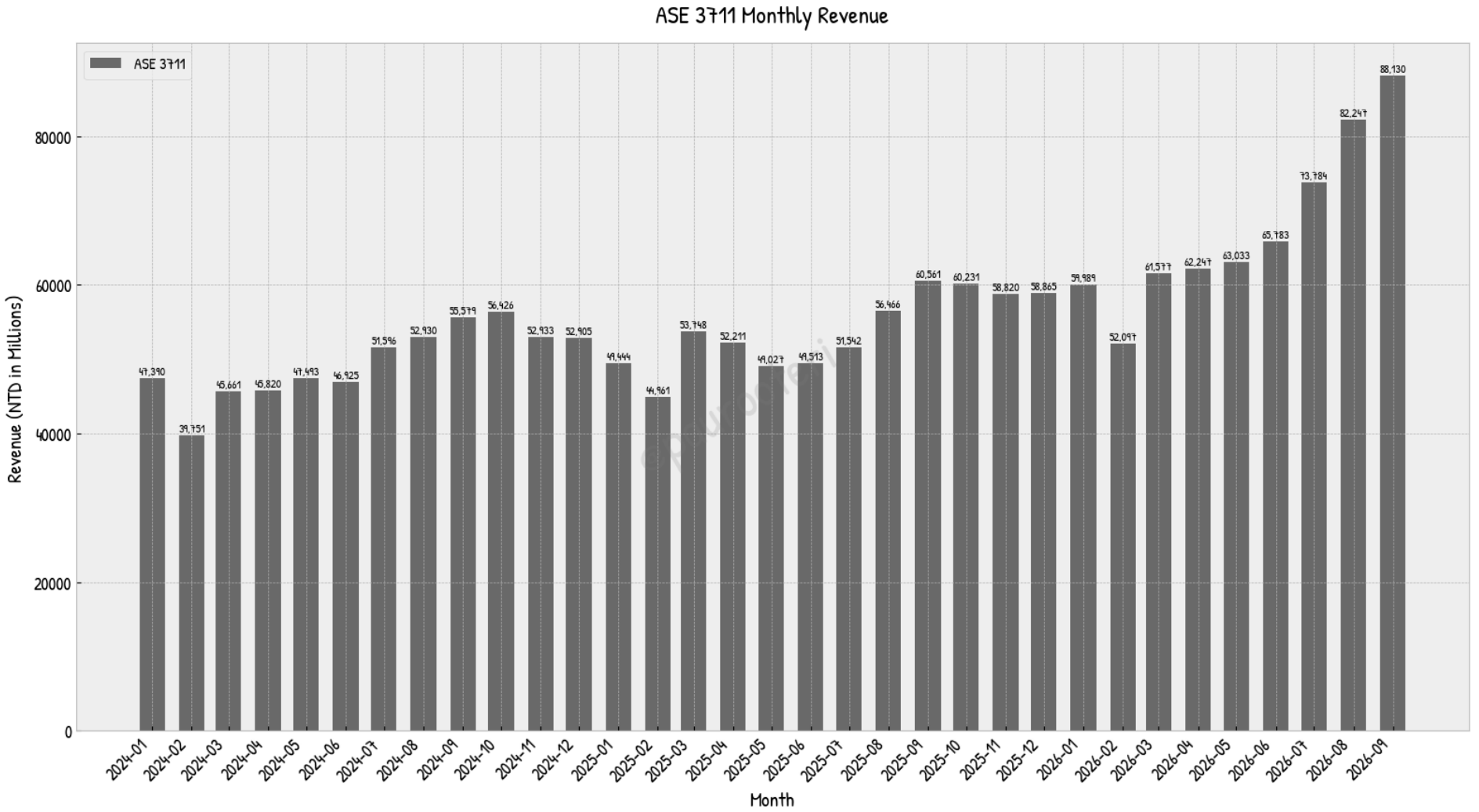


Figure A11. Original monthly consolidated revenue chart from the mapped RAML source post. Units TWD millions, through September2026. Its July73,784+August82,247+September88,130 reconcile244,161. No margins or FoCoS invoices appear on this chart.

Period / boundary	Reported or derived revenue TWD M	Status
July2026 consolidated	73,784	Reported
August2026 consolidated	82,247	Reported
September2026 consolidated	88,130	Reported

Period / boundary	Reported or derived revenue TWD M	Status
Q3 consolidated total	244,161	Reported, unaudited
Q3 ATM	151,485	Reported, unaudited
Q3 non-ATM/eliminations residual	92,676	Derived, net of consolidation

The new quarter is5.177090166413% above the earlier TWD232,142.76M consolidated guide midpoint. ATM is7.218873296219% above the former141,285.76M allocation, while net non-ATM rises2.002047173030%. The model preserves the old within-ATM LEAP/mainstream proportions and product margins. It therefore allocates LEAP TWD37,526.605654M and mainstream ATM TWD113,958.394346M as estimates. The 9% operating-cost convention produces GP TWD51,826.607433M, costs TWD21,974.490000M and OP TWD29,852.117433M. **These profits are modeled, not reported actuals.** Manufacturing costs, labor and depreciation remain in the retained product margins; the source does not identify a fixed-versus-variable opex split.

The incremental estimated Q3 OP is TWD1,994.986233M. No separate current-quarter price increment is added: the previously applied selective ATM pricing phase remains zero in Q3/Q4 2026. Every Q4 2026 onward product path, margin, pricing probability/cohort/overlap and cost assumption is retained. A single aggregate sales beat does not identify the future LEAP cohort, physical output, acceptance, ASP or paid FoCoS scope.

The issuer gives ATM Q3 year-on-year growth as51.0%, consistent with151,485/100,289. A secondary report transposes this to50.1%; the issuer total and percentage govern. Q3 GP/OP, the LEAP/mainstream split, elimination reconciliation and cost absorption remain the next evidence tasks. Historical Q2 cash/debt facts are retained. There is no existing quarterly cash/funding loop or cash-value credit in this model, and no unreported Q3 cash amount is invented.

July 30 management remarks and Q&A, plus April 29 guidance and ramp-cost discussion, were reviewed. The newer call targets approximately double LEAP revenue in 2027 from a 2026 level above $3.5B. It does not disclose a complete LEAP product mix or guarantee the old rumored 20% price increase. Full two-call transcript coverage remains incomplete.

The July 30 prepared remarks and Q&A were reviewed. April 29 financial remarks and selected Q&A were reviewed; complete two-call coverage remains an evidence qualification. Product-level LEAP mix, realized ASPs and margins are undisclosed. The artifact contains the full framework structure while making this source gap explicit.

Equipment procurement and qualified output

The October 6 MOPS disclosure records **TWD1.229287B excluding tax** of equipment purchases by Advanced Semiconductor Engineering, Inc. from unrelated KULICKE AND SOFFA PTE. LTD., accumulated over **July 3-October 6, 2026**. Payment is by wire transfer or letter of credit. This is a cumulative purchase value; the notice does not state that it was all paid in Q4 or that all tools are delivered, accepted or productive. A disclosed aggregate batch is not a machine count. No tool model, site, process routing, delivery schedule or customer acceptance is provided. [A8]

The current execution judgment is **85% probability that the contracted equipment is broadly delivered over its contracted delivery windows**. This is an analyst judgment about procurement execution, not an 85% probability of additional revenue. The equipment supports the existing capacity thesis. Earlier reviewed October 1-2 procurement and October 5 SPIL contracts are distinct disclosed transactions, but their budget overlap is unresolved; the October 5 contracts already supported the existing expansion case without another sales layer.

A bounded capital proxy tests potential overlap before assigning zero incremental output. At the model's retained TWD31.9/US$ assumption, the face value equals **US$38.536M**, approximately **0.367%** of the existing **US$10.5B** capital plan. Genuinely incremental capital above that plan is bounded at **TWD0-1.229287B** until the contract-to-budget mapping is known. This bracket is not added capex guidance or expected cash spending. Applying 85% to the face value gives TWD1.04489395B of illustrative execution exposure only; it does not create an operating revenue forecast.

Saleable output would require delivered tools × throughput × uptime × accepted yield × utilization, followed by invoice content and recognition timing. The disclosure identifies none of these quantities or the portion outside the existing LEAP/mainstream ATM path. **Incremental modeled capacity, revenue and OP therefore remain zero for this filing**, because the bounded capital proxy cannot be converted into separately attributable accepted output and may overlap the existing plan. This is not a rejection of the primary source. Subsequent tool/process, acceptance and budget evidence can change that conclusion. Cash payment, commissioning and useful life also govern cash requirements and later depreciation; purchase value is not expensed a second time in current OP.

ASE | procurement is a capacity input

October 6, 2026 MOPS disclosure | cumulative July 3–October 6 purchases | TWD billions, excluding tax

DISCLOSED PURCHASE VALUE

K&S equipment: 1.229287

ILLUSTRATIVE EXECUTION EXPOSURE

85% × face value = 1.04489395

The 85% delivery judgment concerns execution over contracted delivery windows. The weighted amount is an illustration, not an active cash budget or revenue forecast.

INCREMENT ABOVE EXISTING CAPITAL PLAN: 0–1.229287

At retained 31.9 TWD/USD: US$38.536M, or 0.367% of the existing US$10.5B plan. Budget overlap, tool count, acceptance, throughput and yield are unknown. No separately attributable output or OP increment is modeled.

Source: MOPS, ASE subsidiary filing, 2026-10-06 16:52:06 Taipei | A8

ASE equipment procurement: disclosed purchase value, illustrative execution exposure and overlap limits

Figure A8. Source: October 6 MOPS filing; probability and capital-overlap bracket are analyst judgments. This figure is an explanatory procurement illustration, not a forecast of cash capex, ASE sales or KLIC revenue.

Figure input or calculation	Value	Basis
Cumulative K&S equipment purchases	TWD1.229287B	July 3-October 6, excluding tax
Broad delivery judgment	85%	Analyst execution probability
Illustrative execution exposure	TWD1.04489395B	1.229287 ×85%; no active cash-budget input
Potential amount above existing plan	TWD0-1.229287B	Unresolved overlap bracket
Scale versus existing US$10.5B capital plan	0.367%	Retained TWD31.9/US$; face value US$38.536M

Selective ATM repricing: a small active increment under Rule 16b

The visible lead of the October 12, 02:26 Taipei report (October 11, 14:26 Toronto) attributes tight packaging/test capacity and longer AI testing cycles to industry sources, with reported increases of 10-30%. The rest is gated. This is an attributed industry check, not ASE-confirmed company-wide guidance. The urgent small-customer doubling anecdote is excluded. [A9]

ASTRA assigns **60% probability** that a durable pricing gain reaches a bounded new or repriceable cohort. The conditional premium is **10-30%, central 20%**, applied to **5% of baseline ATM revenue (LEAP plus mainstream, excluding EMS)**. **50% overlap** is removed because existing product margins and invoice-content growth already capture part of favorable economics. The source does not disclose eligible share, contract timing or realized ASE mix; these are explicit company-allocation assumptions.

Recognition phases at 25/50/75/100% in Q1-Q4 2027, 100% through 2028, 50% in 2029 and zero in 2030. The forward start avoids retroactively raising a guided current-quarter result; it is an analyst implementation schedule, not a claim that the article reports 2027 contracts. The fade reflects competition and capacity response. Fixed-price backlog and EMS receive no automatic price increase.

Incremental sales = baseline ATM ×5% eligible ×20% premium ×60% probability ×(1-50% overlap) ×phase. Incremental manufacturing costs consume 20% of those sales and support expense 5%, leaving 75% incremental operating-profit capture. That is a marginal price/cost bridge, not a 75% whole-company margin. Costs cover consumables, labor, maintenance and yield/expedite burden on the existing volume path; manufacturing depreciation stays in baseline product costs. No additional capex or physical-output expansion is implied. Baseline opex is retained before adding the 5% support charge.

At full phase, the expected net price contribution is only **0.30% of baseline ATM revenue**. The 10-30% conditional range therefore translates into 0.15-0.45% of baseline ATM, rather than a sector headline applied to the entire business. Longer qualified test time can support pricing because customers face the cost of delaying expensive AI shipments, but contract resets, qualification and competing capacity still govern realization.

For the active Q4 2027 window, incremental OP is **TWD 1,762.075800M**, lifting the ceiling to **TWD 836.638488**. At otherwise identical assumptions, the 10% and 30% premium bounds give **TWD 832.757806-840.519171**; failure of the new branch returns to the prior **TWD 828.877123** path. These are pricing sensitivities, not separately weighted alternative company valuations.

The narrow 5% cohort and 50% overlap make the increment bounded despite missing ASE-specific terms. Review realized ATM margins alongside material costs, eligible contracts and qualified capacity each quarter; reduce or defer the branch if margins reflect mix alone or if new capacity removes scarcity.

4. Local evidence and calibration packet

Taiwan-IFRS issuer results and monthly revenue establish the reporting baseline. Q2 ATM plus EMS exceeds the consolidated total because of intercompany eliminations. The model assigns eliminations to net EMS to preserve exact consolidation. Q3 revenue 232,142.76M ×21% GP minus 9% opex yields 27,857.1312M OP, matching the 12% midpoint. Product allocation is estimated. July/August are actual sales only; the September residual required by that forecast remains 76,111.76M, with no implied actual profit.

5. Summing thesis score

Category	Points / maximum	Evidence
Core Thesis	16/20	Capacity and qualifications create scarcity, but foundries and competing OSATs retain alternatives.
Cycle	9/10	Reported ATM margin expansion and constrained capacity; not a sole-source monopoly.
TAM	9/10	Full-process packaging, test, optical integration and panel packaging expand work content.
Catalysts	5/5	LEAP ramp, capacity installation and higher structural ATM margin.
Macro	9/10	AI compute and networking complexity sustain packaging demand.
Alpha	3/5	Revenue mix can improve faster than a blended EMS headline suggests.
Risks	-5	Capex/ramp execution, concentrated customers, pricing/cycle risk, funding/dilution and Taiwan/FX exposure.

Sum: 46/60 × 100 = 76.666667%, conventionally rounded to 77%. This advisory thesis score does not replace any portfolio Base override. The separate risk classes below total the displayed deduction; holding status and classification do not waive that deduction.

6. Revenue sources, capture and product margins

Calendar quarters. Preserve 20× forward-four-quarter operating profit. Q3 2026 consolidated and ATM sales are reported, unaudited totals of TWD244,161M and151,485M. Retained within-ATM mix and product margins plus9%opex produce estimated GP TWD51,826.607433M and OP TWD29,852.117433M. The net non-ATM/eliminations residual is92,676M. Shared forward anchors are unchanged; current-quarter realization factors alone match the actual totals. EMS is net of eliminations, so segment totals equal consolidated revenue. Its first-row GM is a reconciliation proxy, not company segment guidance. Q4 assumes ATM GM 30.5% and flat gross EMS sales with net eliminations. Annual LEAP assumptions are US$7.4/11/15/20B for 2027-30 at fixed NT$31.9; only the near-term doubling has management support. Annual mainstream ATM is NT$450/470/495/520B and net EMS NT$330/345/362/380B. Quarter weights 22/24/26/28% are analyst seasonality. LEAP GM 40-44% and mainstream GM 26-28% are unreported product proxies; depreciation stays in manufacturing cost. Opex is 9% in Q3, 9.2% Q4, then 9.5/9/8.7/8.5% for 2027-30. Shares are 4,505.702583M plus 34.931M initial full-conversion shares throughout: a deliberately fully diluted price target proxy, not reported current shares or a forecast of conversion. No cash-value credit is added for bond proceeds. Future conversion-price adjustments remain a sensitivity. The 2030 LEAP endpoint is a conditional demand/capacity bridge:7.4B 2027 ×2.00 productive-volume index ×(20/14.8) process-content/mix index =20B. The 2× volume and 35.135% greater invoice content are analyst assumptions, not disclosed factory output. Physical yield and accepted production, rather than buildings or gross equipment expenditure, govern conversion. No 20% uniform price hike is assumed. The separate A9 branch applies a probability-weighted premium only to a narrow eligible ATM cohort, net of overlap; it does not change these baseline product vectors.

Vector	Actual calibration / source	Forward demand and capture	Margin derivation / constraint
LEAP advanced packaging and test	Q3 estimated LEAP allocation37.526606B TWD; the within-ATM share is retained, not disclosed. Near-term doubling remains the management outlook.	Annual USD 7.4/11/15/20B at 31.9FX. On-substrate, FoCoS, full-process and test all consume this same sales boundary.2030 requires 2.702703×2027 revenue via the stated 2×throughput ×1.351351×content bridge. No exact wafer/package capacity is disclosed to prove it.	40% 2027→44% 2030 product-GM proxy. Cost share 60%→56% includes substrates/materials, labor, depreciation and yield loss. Higher content improves GP only if pricing exceeds additional process cost.
Mainstream ATM	Q3 estimated mainstream allocation113.958394B inside actual151.485B ATM. Product split is not reported.	Annual 450/470/495/520B TWD. Industrial, power, storage and connectivity demand support utilization. These modest growth factors are separate from the LEAP ramp.	26% 2027→28% 2029-30 proxy. Initial margin 24.713% balances 28.5% total ATM guidance; Q4 balances 30.5%. These are not disclosed product GMs.
EMS net of eliminations	Q3 derived non-ATM/eliminations residual92.676B TWD. It is not standalone gross EMS sales.	Annual 330/345/362/380B TWD. Supplier memory costs can raise gross sales without proportionate value added. No extra memory-TAM revenue is allocated to ASE beyond recognized EMS invoices.	Q3 reconciliation 9.337%, later 10% proxy. This must be reopened if component pass-through dominates more of the mix.

Confidence: reported historical totals are HIGH. Source-based forward extrapolations are MEDIUM only where the cited source sizes the relevant period; other product allocations, future capture and margin assumptions are LOW. A source for the historical anchor does not make an unsourced future factor management

guidance. The detailed factors below are the single modeled base case, with uncertain assumptions exposed for review.

V1. LEAP advanced packaging and test (inside ATM)

Sources: A1/2/5/6/7; near-term doubling, later capacity/content scenario. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	35,000.000000	1.072188733	37,526.606	40.000%	15,010.642
Q4 2026	35,000.000000	1.200000000	42,000.000	40.000%	16,800.000
Q1 2027	35,000.000000	1.483805714	51,933.200	40.000%	20,773.280
Q2 2027	35,000.000000	1.618697143	56,654.400	40.000%	22,661.760
Q3 2027	35,000.000000	1.753588571	61,375.600	40.000%	24,550.240
Q4 2027	35,000.000000	1.888480000	66,096.800	40.000%	26,438.720
Q1 2028	35,000.000000	2.205657143	77,198.000	42.000%	32,423.160
Q2 2028	35,000.000000	2.406171429	84,216.000	42.000%	35,370.720
Q3 2028	35,000.000000	2.606685714	91,234.000	42.000%	38,318.280
Q4 2028	35,000.000000	2.807200000	98,252.000	42.000%	41,265.840
Q1 2029	35,000.000000	3.007714286	105,270.000	43.000%	45,266.100
Q2 2029	35,000.000000	3.281142857	114,840.000	43.000%	49,381.200
Q3 2029	35,000.000000	3.554571429	124,410.000	43.000%	53,496.300
Q4 2029	35,000.000000	3.828000000	133,980.000	43.000%	57,611.400
Q1 2030	35,000.000000	4.010285714	140,360.000	44.000%	61,758.400
Q2 2030	35,000.000000	4.374857143	153,120.000	44.000%	67,372.800
Q3 2030	35,000.000000	4.739428571	165,880.000	44.000%	72,987.200
Q4 2030	35,000.000000	5.104000000	178,640.000	44.000%	78,601.600

V2. Mainstream ATM excluding LEAP

Sources: A1/2; mainstream demand and utilization. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	106,285.760000	1.072188733	113,958.394	24.713%	28,162.583
Q4 2026	106,285.760000	1.063171586	113,000.000	26.969%	30,475.000
Q1 2027	106,285.760000	0.931451212	99,000.000	26.000%	25,740.000
Q2 2027	106,285.760000	1.016128595	108,000.000	26.000%	28,080.000
Q3 2027	106,285.760000	1.100805978	117,000.000	26.000%	30,420.000
Q4 2027	106,285.760000	1.185483361	126,000.000	26.000%	32,760.000
Q1 2028	106,285.760000	0.972849044	103,400.000	27.000%	27,918.000
Q2 2028	106,285.760000	1.061289866	112,800.000	27.000%	30,456.000
Q3 2028	106,285.760000	1.149730688	122,200.000	27.000%	32,994.000
Q4 2028	106,285.760000	1.238171510	131,600.000	27.000%	35,532.000
Q1 2029	106,285.760000	1.024596334	108,900.000	28.000%	30,492.000
Q2 2029	106,285.760000	1.117741455	118,800.000	28.000%	33,264.000
Q3 2029	106,285.760000	1.210886576	128,700.000	28.000%	36,036.000
Q4 2029	106,285.760000	1.304031697	138,600.000	28.000%	38,808.000
Q1 2030	106,285.760000	1.076343623	114,400.000	28.000%	32,032.000
Q2 2030	106,285.760000	1.174193043	124,800.000	28.000%	34,944.000
Q3 2030	106,285.760000	1.272042464	135,200.000	28.000%	37,856.000
Q4 2030	106,285.760000	1.369891884	145,600.000	28.000%	40,768.000

V3. EMS and consolidation eliminations, net

Sources: A1/2/3; EMS guide, net eliminations and price pass-through. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

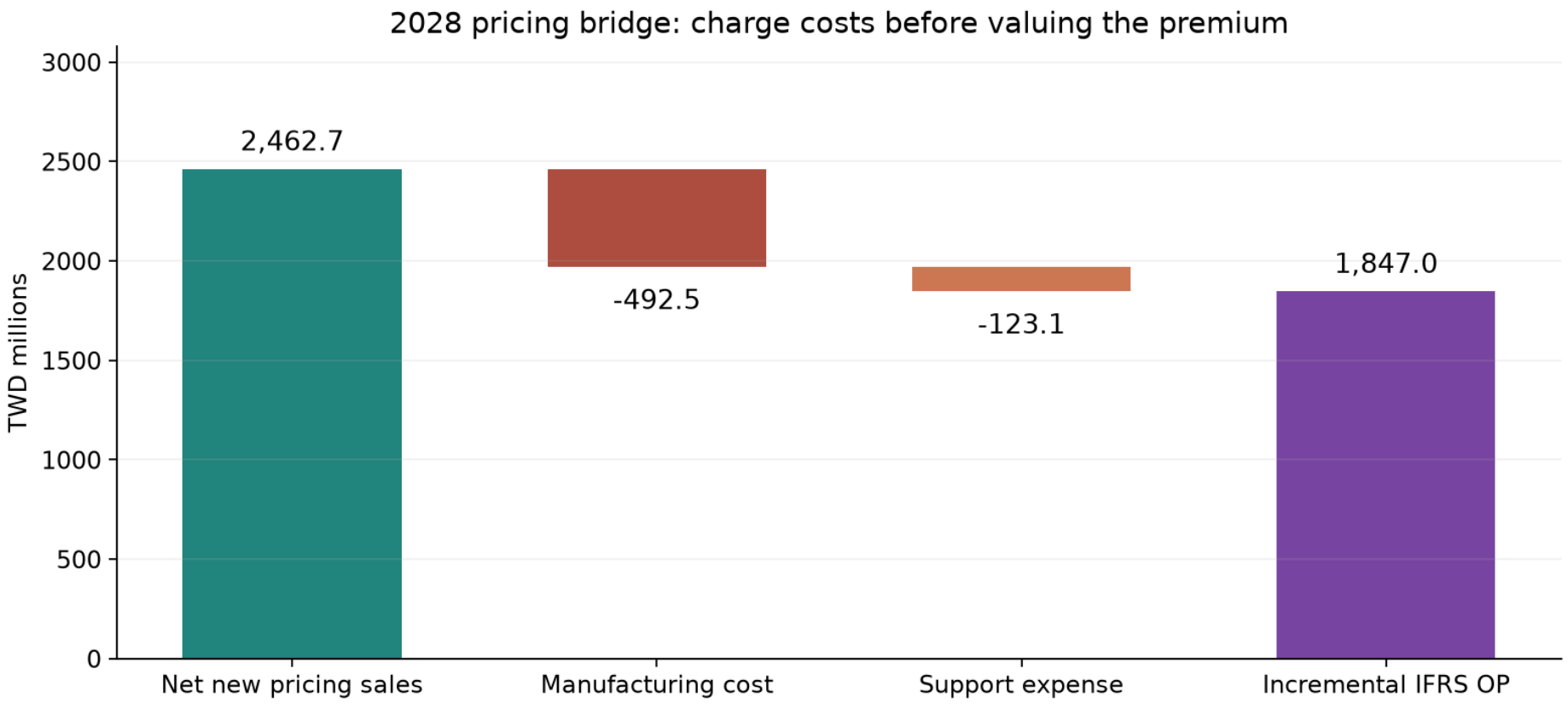
Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	90,857.000000	1.020020472	92,676.000	9.337%	8,653.382
Q4 2026	90,857.000000	0.998272010	90,700.000	10.000%	9,070.000
Q1 2027	90,857.000000	0.799057860	72,600.000	10.000%	7,260.000
Q2 2027	90,857.000000	0.871699484	79,200.000	10.000%	7,920.000
Q3 2027	90,857.000000	0.944341107	85,800.000	10.000%	8,580.000
Q4 2027	90,857.000000	1.016982731	92,400.000	10.000%	9,240.000
Q1 2028	90,857.000000	0.835378672	75,900.000	10.000%	7,590.000
Q2 2028	90,857.000000	0.911322188	82,800.000	10.000%	8,280.000
Q3 2028	90,857.000000	0.987265703	89,700.000	10.000%	8,970.000
Q4 2028	90,857.000000	1.063209219	96,600.000	10.000%	9,660.000
Q1 2029	90,857.000000	0.876542259	79,640.000	10.000%	7,964.000
Q2 2029	90,857.000000	0.956227919	86,880.000	10.000%	8,688.000
Q3 2029	90,857.000000	1.035913578	94,120.000	10.000%	9,412.000
Q4 2029	90,857.000000	1.115599238	101,360.000	10.000%	10,136.000
Q1 2030	90,857.000000	0.920127233	83,600.000	10.000%	8,360.000
Q2 2030	90,857.000000	1.003775163	91,200.000	10.000%	9,120.000
Q3 2030	90,857.000000	1.087423093	98,800.000	10.000%	9,880.000
Q4 2030	90,857.000000	1.171071024	106,400.000	10.000%	10,640.000

V4. Selective pricing overlay within ATM

V1-V3 above retain the pre-pricing physical-volume/content baseline. The overlay below is added once to consolidated revenue and gross profit; it is not a fourth independent end-market TAM. Values are TWD millions. Expected revenue already includes 60% probability and 50% overlap removal.

Quarter	Phase	Baseline ATM	Net pricing revenue	Incremental COGS	Incremental GP	Support opex	Incremental OP
Q3 2026	0%	151,485.000	0.000	0.000	0.000	0.000	0.000

Quarter	Phase	Baseline ATM	Net pricing revenue	Incremental COGS	Incremental GP	Support opex	Incremental OP
Q4 2026	0%	155,000.000	0.000	0.000	0.000	0.000	0.000
Q1 2027	25%	150,933.200	113.200	22.640	90.560	5.660	84.900
Q2 2027	50%	164,654.400	246.982	49.396	197.585	12.349	185.236
Q3 2027	75%	178,375.600	401.345	80.269	321.076	20.067	301.009
Q4 2027	100%	192,096.800	576.290	115.258	461.032	28.815	432.218
Q1 2028	100%	180,598.000	541.794	108.359	433.435	27.090	406.346
Q2 2028	100%	197,016.000	591.048	118.210	472.838	29.552	443.286
Q3 2028	100%	213,434.000	640.302	128.060	512.242	32.015	480.227
Q4 2028	100%	229,852.000	689.556	137.911	551.645	34.478	517.167
Q1 2029	50%	214,170.000	321.255	64.251	257.004	16.063	240.941
Q2 2029	50%	233,640.000	350.460	70.092	280.368	17.523	262.845
Q3 2029	50%	253,110.000	379.665	75.933	303.732	18.983	284.749
Q4 2029	50%	272,580.000	408.870	81.774	327.096	20.444	306.653
Q1 2030	0%	254,760.000	0.000	0.000	0.000	0.000	0.000
Q2 2030	0%	277,920.000	0.000	0.000	0.000	0.000	0.000
Q3 2030	0%	301,080.000	0.000	0.000	0.000	0.000	0.000
Q4 2030	0%	324,240.000	0.000	0.000	0.000	0.000	0.000



ASE 2028 selective-pricing sales, incremental manufacturing/support costs and operating-profit bridge

Figure A9. Calculated from the disclosed ASTRA assumptions, not source-reported financial guidance.

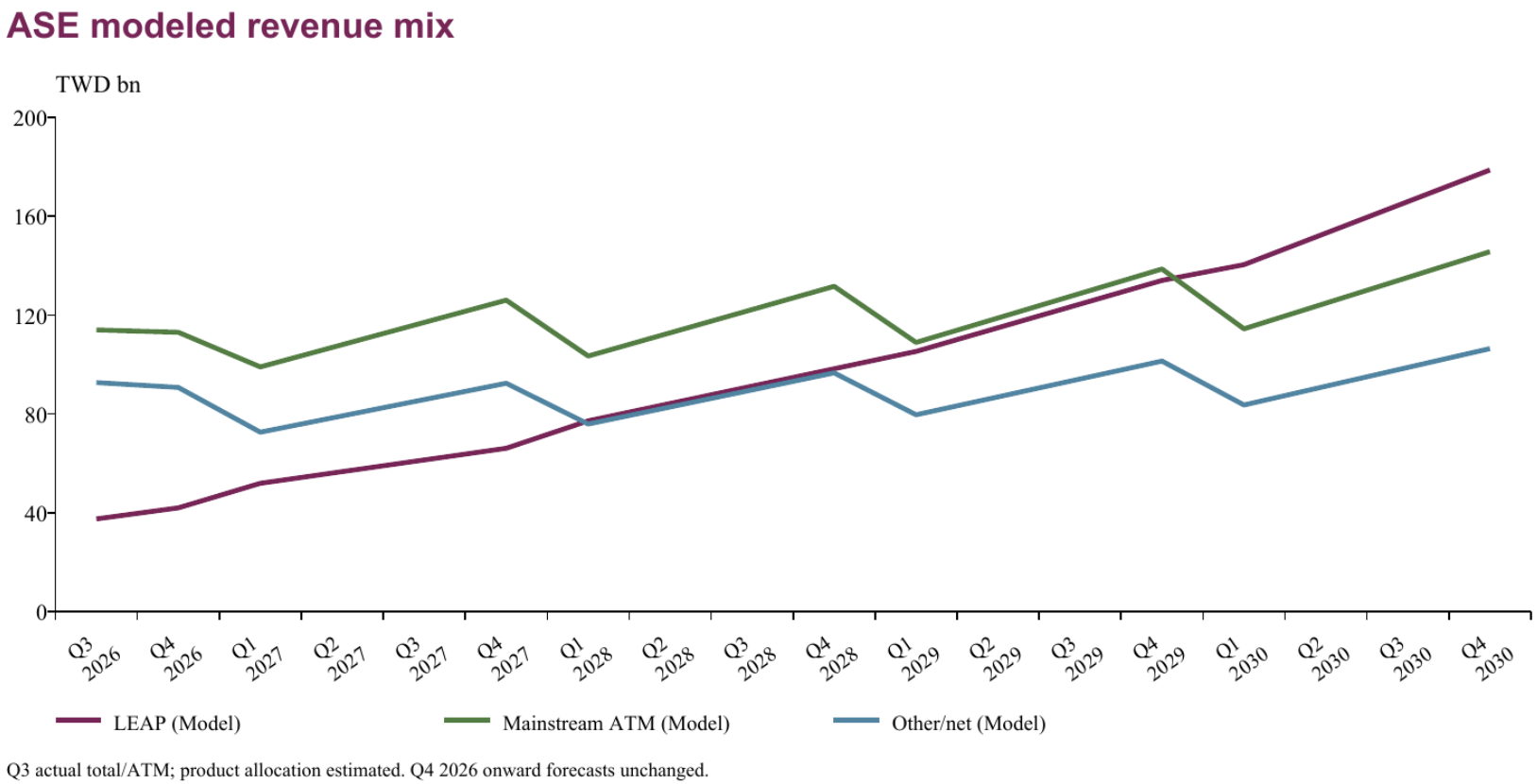


Figure A12. Revenue mix from the synchronized model. Q3 total and ATM are reported, while within-ATM and net residual allocations remain estimates. Units TWD billions. Q4 2026 onward forecasts unchanged.

Quarter	LEAP Model TWD B	Mainstream ATM Model TWD B	Other/net Model TWD B
Q3 2026	37.527	113.958	92.676
Q4 2026	42.000	113.000	90.700
Q1 2027	51.933	99.000	72.600
Q2 2027	56.654	108.000	79.200
Q3 2027	61.376	117.000	85.800
Q4 2027	66.097	126.000	92.400

Quarter	LEAP Model TWD B	Mainstream ATM Model TWD B	Other/net Model TWD B
Q1 2028	77.198	103.400	75.900
Q2 2028	84.216	112.800	82.800
Q3 2028	91.234	122.200	89.700
Q4 2028	98.252	131.600	96.600
Q1 2029	105.270	108.900	79.640
Q2 2029	114.840	118.800	86.880
Q3 2029	124.410	128.700	94.120
Q4 2029	133.980	138.600	101.360
Q1 2030	140.360	114.400	83.600
Q2 2030	153.120	124.800	91.200
Q3 2030	165.880	135.200	98.800
Q4 2030	178.640	145.600	106.400

7. Operating-cost bridge and quarterly operating model

Baseline quarterly costs begin 9%/9.2% of pre-pricing revenue, followed by 9.5% in 2027, 9% in 2028, 8.7% in 2029 and 8.5% in 2030. A9 adds support costs equal to 5% of the incremental pricing revenue. The table shows the resulting effective opex ratio. They cover R&D, corporate and selling functions. Manufacturing depreciation and direct labor are already in product GM. Planned capex is not expensed again in OP, but the resulting cash requirement and subsequent depreciation are material. The model’s lower cost ratio is an execution assumption: building more capacity can increase absolute costs before sales arrive.

Profit basis: IFRS operating profit before financing, tax and non-operating investments. Amounts below are TWD millions.

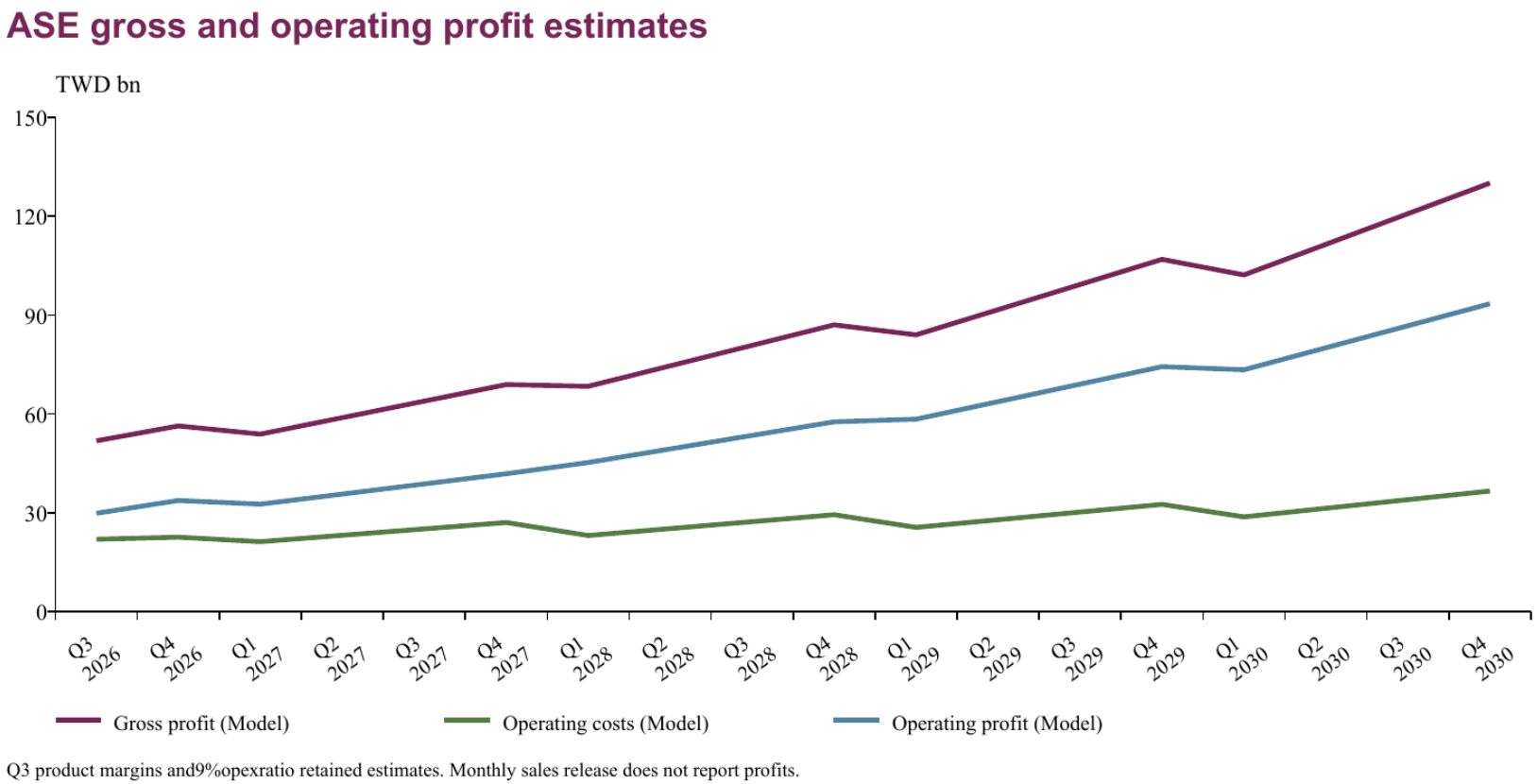


Figure A13. Gross profit, operating costs and operating profit from the current model, TWD billions. All profits remain estimated, including Q3 with reported sales. No additional capex expense or cash-value multiple is introduced.

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q3 2026	Actual sales; Model profits	244,161.000	51,826.607	9.000%	21,974.490	12.226%	29,852.117	Q3 actual sales; margins estimated	LOW mix/profit model
Q4 2026	Model	245,700.000	56,345.000	9.200%	22,604.400	13.732%	33,740.600	ATM structural-margin progression	LOW mix/profit model

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q1 2027	Model	223,646.400	53,863.840	9.498%	21,241.314	14.587%	32,622.526	LEAP capacity installation	LOW mix/profit model
Q2 2027	Model	244,101.382	58,859.345	9.495%	23,178.517	14.617%	35,680.828	Renwu phase 1 timing	LOW mix/profit model
Q3 2027	Model	264,576.945	63,871.316	9.493%	25,116.749	14.648%	38,754.567	2027 qualified production	LOW mix/profit model
Q4 2027	Model	285,073.090	68,899.752	9.491%	27,056.011	14.678%	41,843.742	Renwu phase 2 timing	LOW mix/profit model
Q1 2028	Model	257,039.794	68,364.595	8.992%	23,111.910	17.605%	45,252.686	2028 test/content expansion	LOW mix/profit model
Q2 2028	Model	280,407.048	74,579.558	8.992%	25,212.992	17.605%	49,366.566	New-building completion target	LOW mix/profit model
Q3 2028	Model	303,774.302	80,794.522	8.992%	27,314.075	17.605%	53,480.446	Equipment qualification	LOW mix/profit model
Q4 2028	Model	327,141.556	87,009.485	8.992%	29,415.158	17.605%	57,594.327	2028 production conversion	LOW mix/profit model
Q1 2029	Model	294,131.255	83,979.104	8.696%	25,577.533	19.856%	58,401.571	2029 broader packaging demand	LOW mix/profit model
Q2 2029	Model	320,870.460	91,613.568	8.696%	27,902.763	19.856%	63,710.805	More process content	LOW mix/profit model
Q3 2029	Model	347,609.665	99,248.032	8.696%	30,227.993	19.856%	69,020.039	Advanced test complexity	LOW mix/profit model
Q4 2029	Model	374,348.870	106,882.496	8.696%	32,553.224	19.856%	74,329.273	2029 capacity utilization	LOW mix/profit model
Q1 2030	Model	338,360.000	102,150.400	8.500%	28,760.600	21.690%	73,389.800	2030 capacity/ content case	LOW mix/profit model
Q2 2030	Model	369,120.000	111,436.800	8.500%	31,375.200	21.690%	80,061.600	Continued customer conversion	LOW mix/profit model

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q3 2030	Model	399,880.000	120,723.200	8.500%	33,989.800	21.690%	86,733.400	Mature LEAP mix	LOW mix/profit model
Q4 2030	Model	430,640.000	130,009.600	8.500%	36,604.400	21.690%	93,405.200	Conditional 20B LEAP annual case	LOW mix/profit model

OP = sum of baseline product GP/contribution + A9 incremental GP - retained baseline operating costs - A9 support expense. OP margin = OP ÷ revenue. Displayed numbers are rounded for reading; the workbook calculates from unrounded inputs. Guidance assigned to a future quarter retains Model status because guidance is not an actual result.

8. Multiple, shares and recognition convention

20×forward IFRS OP reflects qualified advanced packaging and test with rising value added. Capital intensity, cyclical utilization and the larger low-margin EMS business constrain the premium. No automatic multiple uplift accompanies the expanded LEAP demand path, and non-operating investment gains do not enter OP.

4,505.702583M latest diluted weighted-average shares +1,000M USD×32.388÷927.20 =34.930975M full bond-conversion shares gives 4,540.633558M. That denominator is used throughout. Full conversion is a valuation assumption; the bond remains debt until conversion. Proceeds do not receive a separate cash-value credit. Conversion-price resets, employee awards and later funding can change dilution. Q2 cash/current financial assets 107.4B TWD and interest-bearing debt 306.2B demonstrate substantial financing exposure; undrawn facilities are borrowing capacity, not cash. [A1/A2/A4]

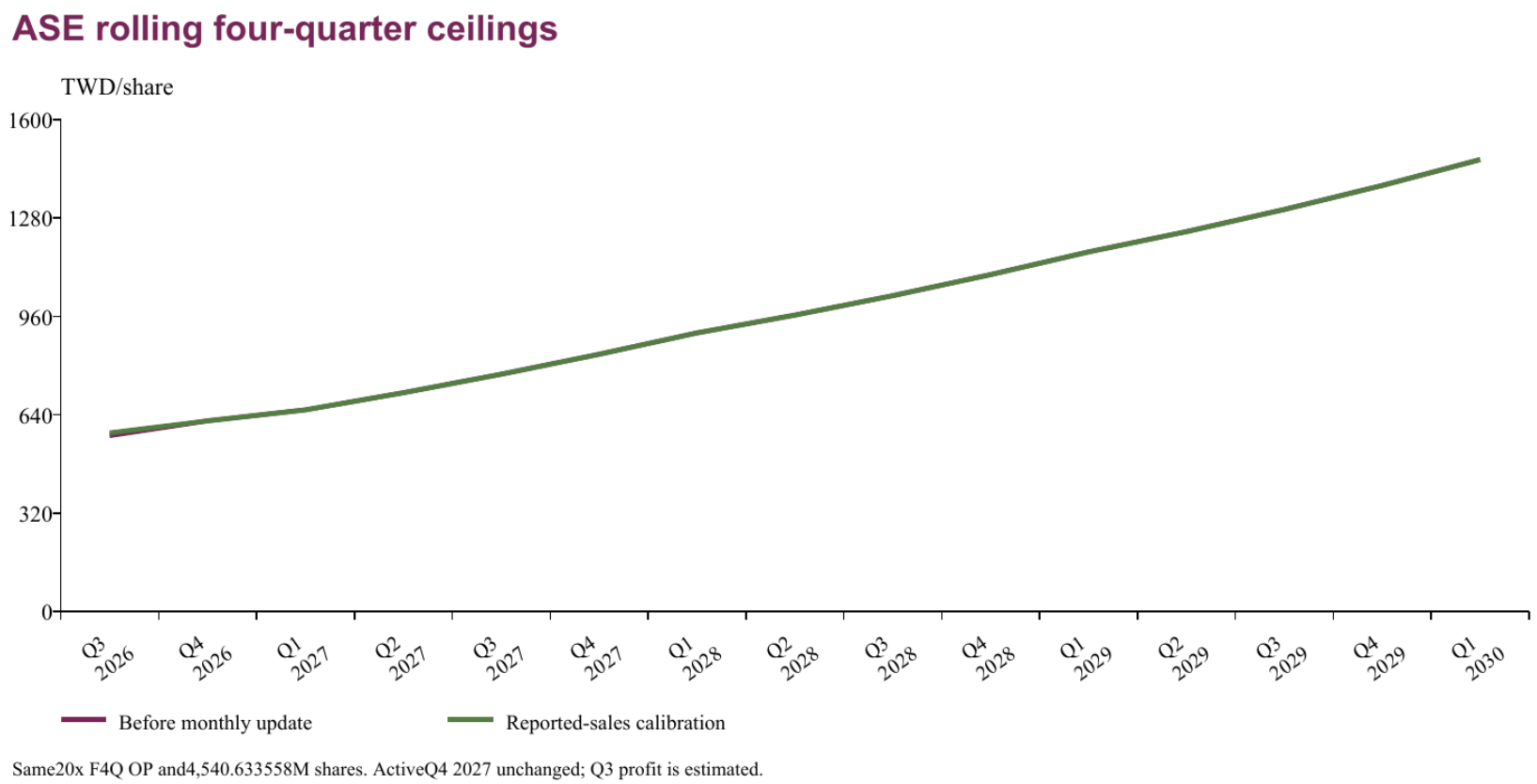
The valuation denominator is the price target quarter’s diluted-share proxy. It is not a historical EPS weighted-average assumption disguised as a fully known future capitalization. Unit factor: 1; this converts the model units to TWD per share. No cash balance, enterprise-value adjustment or after-tax earnings substitution is added to the operating-profit price target.

9. Complete rolling four-quarter price target derivation

Price target quarter	Exact earnings window	Forward 4Q OP	Multiple	Diluted shares M	Price target TWD	Key inflection
Q3 2026	Q3 2026 to Q2 2027	131,896.071558	20x	4,540.633558	580.958890	Q3 actual sales; margins estimated
Q4 2026	Q4 2026 to Q3 2027	140,798.520950	20x	4,540.633558	620.171257	ATM structural-margin progression
Q1 2027	Q1 2027 to Q4 2027	148,901.662750	20x	4,540.633558	655.862936	LEAP capacity installation
Q2 2027	Q2 2027 to Q1 2028	161,531.822325	20x	4,540.633558	711.494642	Renwu phase 1 timing
Q3 2027	Q3 2027 to Q2 2028	175,217.560125	20x	4,540.633558	771.775823	2027 qualified production
Q4 2027	Q4 2027 to Q3 2028	189,943.439800	20x	4,540.633558	836.638488	Renwu phase 2 timing
Q1 2028	Q1 2028 to Q4 2028	205,694.025000	20x	4,540.633558	906.014645	2028 test/content expansion
Q2 2028	Q2 2028 to Q1 2029	218,842.910750	20x	4,540.633558	963.931169	New-building completion target
Q3 2028	Q3 2028 to Q2 2029	233,187.149750	20x	4,540.633558	1,027.112832	Equipment qualification
Q4 2028	Q4 2028 to Q3 2029	248,726.742000	20x	4,540.633558	1,095.559634	2028 production conversion
Q1 2029	Q1 2029 to Q4 2029	265,461.687500	20x	4,540.633558	1,169.271575	2029 broader packaging demand
Q2 2029	Q2 2029 to Q1 2030	280,449.916250	20x	4,540.633558	1,235.289801	More process content
Q3 2029	Q3 2029 to Q2 2030	296,800.711250	20x	4,540.633558	1,307.309685	Advanced test complexity
Q4 2029	Q4 2029 to Q3 2030	314,514.072500	20x	4,540.633558	1,385.331225	2029 capacity utilization
Q1 2030	Q1 2030 to Q4 2030	333,590.000000	20x	4,540.633558	1,469.354423	2030 capacity/content case

- **Q3 2026:** (29,852.117433 + 33,740.600000 + 32,622.525925 + 35,680.828200) x20 x1 /4,540.633558 = **580.958890 TWD/share**. Window: Q3 2026-Q2 2027. Q3 revenue is reported; profits remain estimates.
- **Q4 2026:** (33,740.600000 + 32,622.525925 + 35,680.828200 + 38,754.566825) ×20 ×1 ÷4,540.633558 = **620.171257 TWD/share**. Window:Q4 2026-Q3 2027.
- **Q1 2027:** (32,622.525925 + 35,680.828200 + 38,754.566825 + 41,843.741800) ×20 ×1 ÷4,540.633558 = **655.862936 TWD/share**. Window:Q1 2027-Q4 2027.
- **Q2 2027:** (35,680.828200 + 38,754.566825 + 41,843.741800 + 45,252.685500) ×20 ×1 ÷4,540.633558 = **711.494642 TWD/share**. Window:Q2 2027-Q1 2028.
- **Q3 2027:** (38,754.566825 + 41,843.741800 + 45,252.685500 + 49,366.566000) ×20 ×1 ÷4,540.633558 = **771.775823 TWD/share**. Window:Q3 2027-Q2 2028.
- **Q4 2027:** (41,843.741800 + 45,252.685500 + 49,366.566000 + 53,480.446500) ×20 ×1 ÷4,540.633558 = **836.638488 TWD/share**. Window:Q4 2027-Q3 2028.

- **Q1 2028:** (45,252.685500 + 49,366.566000 + 53,480.446500 + 57,594.327000) ×20 ×1 ÷4,540.633558 = **906.014645 TWD/share**. Window:Q1 2028-Q4 2028.
- **Q2 2028:** (49,366.566000 + 53,480.446500 + 57,594.327000 + 58,401.571250) ×20 ×1 ÷4,540.633558 = **963.931169 TWD/share**. Window:Q2 2028-Q1 2029.
- **Q3 2028:** (53,480.446500 + 57,594.327000 + 58,401.571250 + 63,710.805000) ×20 ×1 ÷4,540.633558 = **1,027.112832 TWD/share**. Window:Q3 2028-Q2 2029.
- **Q4 2028:** (57,594.327000 + 58,401.571250 + 63,710.805000 + 69,020.038750) ×20 ×1 ÷4,540.633558 = **1,095.559634 TWD/share**. Window:Q4 2028-Q3 2029.
- **Q1 2029:** (58,401.571250 + 63,710.805000 + 69,020.038750 + 74,329.272500) ×20 ×1 ÷4,540.633558 = **1,169.271575 TWD/share**. Window:Q1 2029-Q4 2029.
- **Q2 2029:** (63,710.805000 + 69,020.038750 + 74,329.272500 + 73,389.800000) ×20 ×1 ÷4,540.633558 = **1,235.289801 TWD/share**. Window:Q2 2029-Q1 2030.
- **Q3 2029:** (69,020.038750 + 74,329.272500 + 73,389.800000 + 80,061.600000) ×20 ×1 ÷4,540.633558 = **1,307.309685 TWD/share**. Window:Q3 2029-Q2 2030.
- **Q4 2029:** (74,329.272500 + 73,389.800000 + 80,061.600000 + 86,733.400000) ×20 ×1 ÷4,540.633558 = **1,385.331225 TWD/share**. Window:Q4 2029-Q3 2030.
- **Q1 2030:** (73,389.800000 + 80,061.600000 + 86,733.400000 + 93,405.200000) ×20 ×1 ÷4,540.633558 = **1,469.354423 TWD/share**. Window:Q1 2030-Q4 2030.



ASE exact quarterly ceiling path before and after monthly sales calibration

Figure A10. Exact model before/after the reported-sales calibration. Only the Q3 2026 earnings window changes; the active Q4 2027 ceiling is unchanged. Product mix and profits are estimates. Method and denominator are preserved.

10. Annual trajectory cross-check

Calendar year	Revenue	GP / contribution	Opex	OP	OP margin
2027	1,017,397.817	245,494.254	96,592.591	148,901.663	14.636%
2028	1,168,362.700	310,748.160	105,054.135	205,694.025	17.605%
2029	1,336,960.250	381,723.200	116,261.512	265,461.688	19.856%
2030	1,538,000.000	464,320.000	130,730.000	333,590.000	21.690%

These are calendar-year sums of the quarterly path, not extra revenue and not the live valuation method. For non-December fiscal reporters, fiscal guidance and these calendar totals must be aligned using the period mapping above; a four-quarter run rate is not automatically the same fiscal year. Outer-year outputs are conditional TAM/capture trajectories, not verified management sales targets.

2027-30 consolidated revenue is 1,017.397817B/1,168.362700B/1,336.960250B/1,538.000000B TWD, including the narrow selective-pricing overlay. The baseline LEAP, mainstream ATM and net EMS volume/content assumptions are unchanged. The 20B USD LEAP endpoint is a forward scenario with a disclosed throughput/content requirement. The new Renwu and Kaohsiung projects support a capacity path; published site or cluster output value is not added to the same consolidated sales. Qualification delay should move recognition and expense absorption, even if the long-term demand envelope grows.

11. Q3 2027 dilution sensitivity

Denominator sensitivity	Q3 2027 shares M	Q3 2027 target TWD	Target change
-5% shares	4,313.601880	812.395603	+5.26%
+0% shares	4,540.633558	771.775823	+0.00%
+5% shares	4,767.665236	735.024594	-4.76%
+10% shares	4,994.696914	701.614385	-9.09%

The table retains the Q3 2027 earnings window, refreshed for the same pricing branch, to isolate denominator sensitivity. The active Q4 2027 target is TWD836.638488.

This isolates denominator uncertainty; it is not a second bull/bear operating case or permission to remove known dilution. The base case remains the explicit share path in the rolling table.

12. Programme and probability ledger

Programme / signal	Source and probability	Amount / unit base	Treatment in this model	Escalation / recognition
LEAP2027 doubling	Management plus execution plan; 80% confidence	Illustrative 3.7B 2026×2	7.4B embedded once	Actual 2026 base and 2027 qualified shipments replace the assumption.
LEAP2030 expansion	Analyst conditional capture; 50% confidence	2×productive volume×1.351351 content	20B inside existing vector	New capacity/output disclosures must support throughput and process mix.
Existing pricing environment	Management favorable terms; 80% qualitative confidence retained	Favorable product margin/content already in baseline	Existing economics retained; no separate application of this prior confidence	Realized product GP after inflation.
A9 selective new/repriceable ATM cohort	Industry report; 60% durable realization probability	10-30% premium,central 20%; 5% baseline ATM; 50% overlap	Phase 25/50/75/100% in 2027, 100% in 2028, 50% in 2029,zero in 2030. Costs 20% + support 5% of incremental sales	Contract resets, eligible mix, accepted output, margins after costs; source lead only, scope/timing assumptions.
Panel and optical integration	Management development; 60% timing confidence	No defensible stand-alone invoice forecast	Within LEAP; not additional TAM revenue	Accepted production and product economics.
Renwu phases	Issuer construction plan;75% schedule confidence	April/October 2027 targeted operations	Capacity evidence only; cluster output is not ASE sales	Equipment installation, customer qualification and achieved utilization.
K&S equipment procurement	October 6 primary filing; 85% broad delivery confidence	TWD1.229287B cumulative July 3-October 6, excluding tax	Capital-overlap bracket TWD0-1.229287B; zero additional output/OP because accepted-output and budget mapping are unavailable	Tool/process routing, delivery and acceptance dates, good-unit yield, utilization, useful life and overlap with the existing US$10.5B plan.

Probabilities are analyst judgments under the framework signal hierarchy. Embedded revenue is not multiplied or added a second time. Where no defensible amount or unit base exists, the opportunity remains explicitly unquantified; confidence alone cannot create a revenue number. New evidence can enlarge the TAM, improve capture or move recognition earlier, but each requires an identifiable model input.

13. Scored risk classes

Risk class	Scored deduction	Mechanism and consequence
Capacity and return execution	-1	Buildings, equipment, staff and depreciation precede productive output; yield shortfalls impair returns.
Customer and foundry concentration	-1	Leading customers control packaging allocation and can internalize more work.
Pricing and cycles	-1	Component pass-through can inflate revenue while competition or underutilization compresses GP.
Funding and dilution	-1	Heavy capex and negative free cash flow require financing; convertibles can dilute shares.
Taiwan and foreign exchange	-1	Physical concentration and currency movement affect availability and reported margins.

14. Dated catalysts, counters and escalation

1. **2026-11-15 monitoring date - Q3 report** Reconcile reported consolidated/ATM sales with Q3 GP/OP, product mix and cash. September sales are now reported; do not continue treating them as an open gap. Reconcile the K&S procurement to the existing capital plan and obtain delivery, acceptance and productive-output evidence; the October 6 purchase-window end is not a commissioning deadline.
2. **2026-12-31 - ATM margin** Test the >30% exit expectation against realized mix and costs. Obtain eligible contract/reset mix and check whether reported price increases survive manufacturing and support costs; revisit the A9 cohort, overlap and 2027 start rather than automatically carrying it forward.
3. **2027-03-31 - Panel production** Qualification, paid orders and yield establish commercial scale.
4. **2027-04-30 and 2027-10-31 - Renwu phase milestones** Commissioning is necessary; customer acceptance establishes productive capacity.
5. **2028-06-30 - Kaohsiung buildings** Completion must be followed by equipment, yield and shipments.
6. **Every report through 2030 - Capital returns** Compare OP with cash spending, debt and actual dilution; reopen the 20B scenario if conversion lags.

15. Cross-book monitoring

ASE packages and tests silicon sold by other portfolio businesses. LEAP, full-process CoWoS, FoCoS, advanced test and emerging optical/panel packaging overlap: they are subdivisions of the LEAP opportunity, not five independent additive TAMs. A potential EMIB service route broadens addressable work but has no separate forecast revenue here. MRVL/AVGO/AMD drive silicon demand; greater chip complexity can raise ASE invoice content even without identical unit growth. BOE's future glass work and panel equipment suppliers address other parts of the stack. A new materials platform can change cost and yield without proving ASE loses its qualified assembly/test role.

16. Source register and model limits

A1 - Q2 company results, Form 6-K (2026-07-30). The filed results provide consolidated and segment margins, operating profit and diluted shares. Financial investment and currency gains are excluded from this operating-profit model.

A2 - Q2 management call (2026-07-30). Q3 guide: revenue +21-22% QoQ, GM 20.5-21.5%, OP margin 11.5-12.5%; ATM revenue +11-13% and GM 28-29%. Capex rises to $10.5B. EMS growth includes memory-price pass-through. Panel production is expected in Q1 2027; CPO detail remains pending.

A3 - August company revenue filing (2026-09-09). Monthly sales provide revenue, not monthly profit. July plus August totaled TWD156,031M. The former September guide-derived residual TWD76,111.76M is historical and is now superseded by issuer-reported TWD88,130M in A10. The current Q3 total is TWD244,161M; quarterly mix and profits remain estimates.

A4 - Convertible bond offering memorandum (2026-08-10). The $1B zero-coupon bond has initial conversion price NT$927.20 and fixed FX NT$32.388/US$. Initial full conversion implies 34.931M additional common shares.

A5 - Prior management call (2026-04-29). Earlier guidance highlighted ramp costs before capacity reaches steady-state utilization.

A6 - Renwu testing cluster investment (2026-04-10). Issuer release describes a joint testing cluster with phases planned for April and October 2027. Published projected output value is a project/cluster indication, not additive ASE consolidated revenue.

A7 - New Kaohsiung facilities (2026-03-11). Two buildings planned for Q2 2028 completion support packaging and test capacity; construction is not customer-qualified output.

A8 - October 6 equipment procurement filing (2026-10-06, 16:52:06 Taipei). Original MOPS disclosure for ASE subsidiary purchases from KULICKE AND SOFFA PTE. LTD.; July 3-October 6 cumulative amount TWD1,229,287 thousand excluding tax. Local original HTML/text retained and hash-verified. Equipment count, site, acceptance, capacity and incremental-budget mapping are undisclosed.

A9 - Industry reporting on packaging/test prices (2026-10-12, 02:26:15 Taipei; 2026-10-11, 14:26:15 Toronto). The accessible lead reports capacity pressure and industry price increases of 10-30%. Full article gated. Source-linked Grok BQYV posts 35/36 share one underlying event. Model probability, eligible scope, overlap, phased realization and costs are ASTRA assumptions; urgent-order doubling is excluded.

A10 - September and Q3 issuer monthly sales release (2026-10-12,03:00 ET). ASE Technology Holding via PRNewswire. Unaudited consolidated revenue88,130M in September and244,161M in Q3; ATM52,674M and151,485M. UnitTWD millions. Original full HTML and tables retained, SHA256f604203afc366ee466d268f0835df5543536ffc2c9cc01df8074a4b0e0b189dc. Q3 ATM growth is51.0%YoY. Gross EMS revenue/eliminations separately, GP, OP and LEAP product mix are not provided. The mapped Grok RAML/R0 source post2109608568643317923 includes the original monthly consolidated chart. Independently retrieved issuer totals govern over the source commentary.

LEAP quarterly mix, product GM, long-term capacity conversion and seasonality are LOW-confidence estimates. The published call transcript contains apparent unit/transcription errors; no disputed full-process dollar figure is used. April call was reviewed selectively. The full-conversion denominator is a scenario proxy; no claim is made that all bonds are currently dilutive under accounting rules.

Workbook: v3_2_master_portfolio.xlsx, model tab 3711_TW_Q_Model. It owns portfolio ranking, prices and return calculations; those are intentionally outside this artifact. **Current artifact path:** G:\My Drive\1amInvesting\Artifacts\3711TW_ASE_Technology_DeepDive_v4.3.1.md. **Original snapshot:** versions/originals/3711TW_ASE_Technology_DeepDive_v4.3.1.md. Superseded history is retained separately in VERSION_LOG, not in this current-state model.